Climate-Driven Vulnerability Assessment of Hamilton's Urban Forest:

A Ward-Level Resilience Framework for 2030, 2050, and 2080

Authors: Susana Carolina Romero Aparicio, Su Myat Aung, Tessy Ramirez, Yi Xin

Affiliations: University of Niagara Falls Canada

BACKGROUND

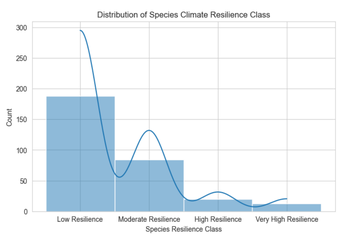
- **Impact:** Environmental - Economic - Social
- **Inventory:** 275,156 city-owned trees · 15 wards · 98.4% species-matched (exact + genus fallback)
- **Climate:** Baseline = 32.82°C (1991–2020) → projected +3.11°C by 2030, +3.76°C by 2050, +4.47°C by 2080 under SSP2-4.5
- **Goal:** Monitor and Improve Resiliency

Urban Forests are KEY Climate Assets BUT increasingly Vulnerable to Climate Stress.

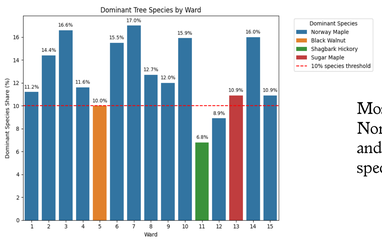
RESEARCH QUESTIONS

- How vulnerable is Hamilton's current urban forest to future climate change? *(Climate Exposure)*
- Does the current species composition provide sufficient resilience to future climate conditions? *(Species Resilience)*
- Which wards and dominant species exhibit the highest climate vulnerability? *(Spatial Risk Concentration)*

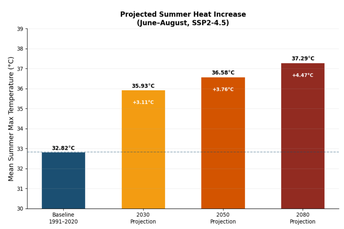
HAMILTON'S URBAN FOREST TODAY



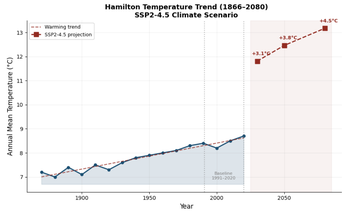
Most tree species exhibit low climate resilience, highlighting vulnerability in the current species mix.



Most wards are dominated by Norway Maple, with Wards 5, 11, and 13 showing alternative dominant species.



Hamilton's summer heat is projected to rise sharply by 2030, 2050, and 2080, increasing climate stress on the city's urban forest.



Hamilton has warmed steadily since the late 19th century and is projected to continue warming through 2080 under the SSP2-4.5 scenario.

VULNERABILITY SCORE FRAMEWORK

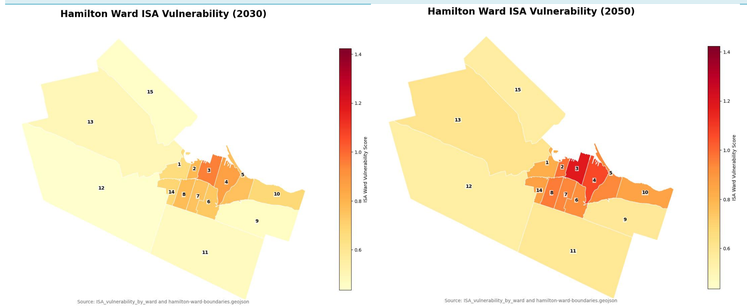
Vulnerability Score = Exposure + Sensitivity - Adaptive Capacity

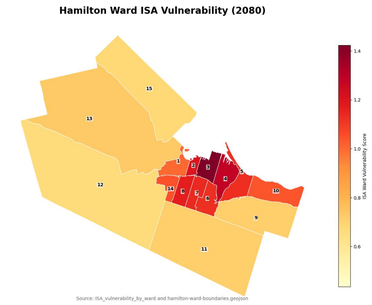
(IPCC AR5 Vulnerability Conceptual framework)

Exposure	Magnitude of climate stress experienced by trees and urban areas	Extreme Heat Intensity (**ΔTXx**): change in annual maximum daily temperature during summer. **Heat index = Σ (Population_category × ISA Impervious Surface Weight)**	Spatial scale: City & Ward Levels
Sensitivity (1 – Resilience score)	Species-level susceptibility to climate stress	**Heat** (AHS Temp >30°C), **Hydroclimate** (Drought + Saturated Soil Tolerance), **Pollution** Tolerance, **Climate** Suitability	Species-Level susceptibility to climate stress
Adaptive Capacity	Ecosystem ability to cope with environmental stress	**DBH** (Diameter at Breast Height). Larger trees exhibit greater resistance through deeper root systems and higher biomass.	Individual Tree Level

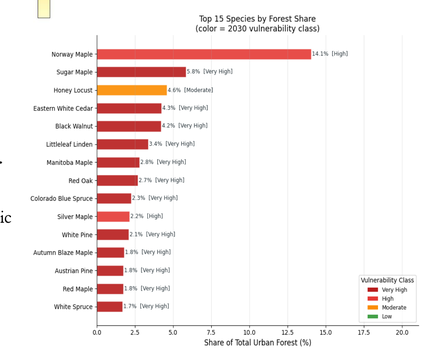
Higher Vulnerability Scores indicate greater climate risk to the urban forest.

All the datasets used in this analysis originates from well-established and widely recognized government-affiliated sources.



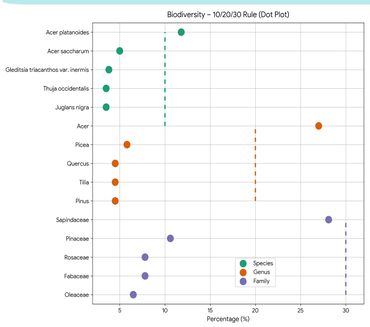


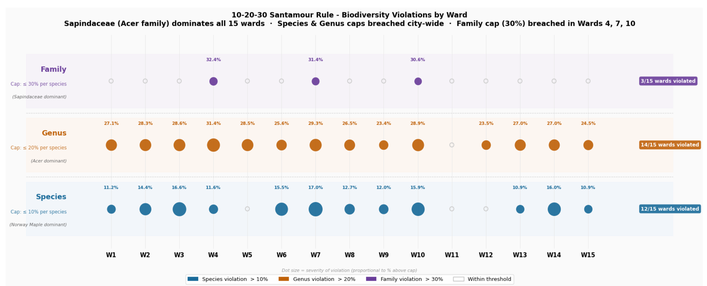
- Norway maple, a dominant urban species, contributes significantly to vulnerability due to its high proportion within the urban forest.
- Among the most vulnerable species, Norway maple stands out due to its relatively high proportion in the urban forest.



- Wards 3, 4, and 2 are most vulnerable. Ward 3 shows the highest projected vulnerability by 2080 under climate change scenarios, necessitating strategic planning prioritization.

SANTAMOUR 10/20/30 RULE FOR URBAN FOREST BIODIVERSITY



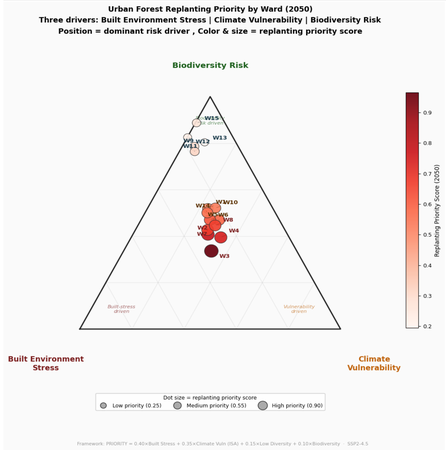


- The Santamour guideline recommends that no single species exceed 10%, no genus exceed 20%, and no family exceed 30% of the urban forest composition.
- Points located to the right of the dashed thresholds indicate biodiversity dominance exceeding the 10/20/30 guideline.
- City-Level: Acer platanoides (Norway maple) exceeds the 10% species threshold, and the Acer genus exceeds the 20% genus threshold. The Sapindaceae family is approaching the 30% family dominance threshold. Ward 4 is the only ward with two species violating the rule.
- Dominance of Acer platanoides increases systemic risk because monoculture urban forests are more susceptible to pest outbreaks and climate stress.

URBAN FOREST REPLANTING PRIORITIES

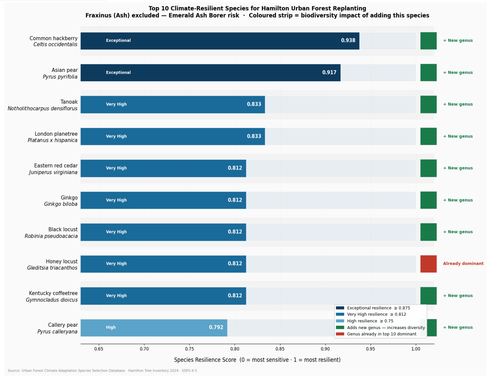
Tree Priority Replanting Rank Framework

[Built Environment Stress + Ward Vulnerability Score + Low Diversity Pressure + Biodiversity Pressure]



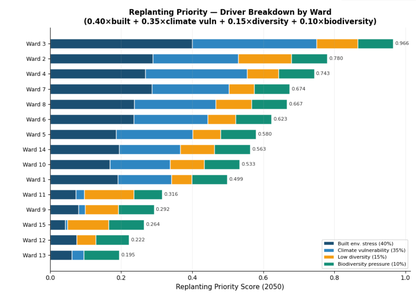
Ward position reflects the dominant driver of 2050 replanting priority.

- Wards 9, 11, 12, 13, and 15 cluster toward biodiversity risk, indicating higher species dominance and lower diversity.
- Wards 2, 3, 4, and 7 show mixed drivers, where built-environment stress and biodiversity pressure combine.
- Ward 3 has the highest replanting priority (0.966) due to intense built-environment heat exposure.



Replanting recommendations prioritize climate-resilient species for vulnerable wards.

- Common hackberry (0.938) and Asian pear (0.917) show the highest resilience.
- 9 of 10 species belong to new genera, reducing reliance on Acer.
- Fraxinus (ash) excluded due to *Emerald Ash Borer risk*.
- Supports climate resilience and biodiversity diversification.



- **Ward 3** emerges as the most critical hotspot, combining low resilience with the highest replanting priority. Wards 6, 7, and 8 show high planting pressure despite relatively stronger resilience, indicating expansion pressure rather than immediate canopy decline.

KEY FINDINGS AND RECOMMENDATIONS

Highest vulnerability:

- ✓ Prioritize climate adaptation in high-risk wards.
- ✓ Ward 3 ranks highest across all climate horizons, driven by the city's highest impervious-surface heat exposure (ISA index = 1.0 normalized).

Biodiversity risk:

- ✓ Reduce dependence on dominant maple species.
- ✓ **Norway Maple** represents 11.9%, of the urban forest, exceeding the 10% Santamour cap.
- ✓ **Acer** genus exceeds the 20% limit in 14 of 15 wards (Ward 4: 31%, Ward 3 & 7: ~29%)

Safest wards: Wards 9, 11, 12, 13, 15 - lowest vulnerability, lowest built share, lowest road stress

Ash risk: 9,170 Fraxinus trees (3.3% of the inventory) are highly vulnerable to Emerald Ash Borer, posing a significant future canopy loss risk. Fraxinus species are excluded from recommended replanting.

DATA SOURCES: **City of Hamilton Open Data Portal** (tree inventory, land use, road network) | **Global Biodiversity Information Facility GBIF Backbone Taxonomy** (species classification validation) | **Metro Vancouver Urban Forest Climate Adaptation Species Selection Database** (species resilience scores)|**CanDCS-M6 / Canadian Centre for Climate Services** (climate scenario projections) | **Ontario Climate Station** Records (historical climate data)

- **REFERENCES:** Santamour, F. S. Jr. (1990) - Urban forest 10-20-30 diversity rule, IPCC AR5 WGII Chapter 19 (2014) - Vulnerability assessment framework, Stewart, City of Hamilton Urban Forest Strategy (2020) - Policy alignment, I. D. & Oke, T. R. (2012) - Local Climate Zones (LCZ) framework, Sjöman, H., Hirons, A. D. & Bassuk, N. L. (2015) - Urban tree drought tolerance